# Encoding Music Score Data for Emotional Expression Prediction in Western Classical Music Pieces

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## Abstract

Music's ability to evoke emotions has long been recognised, with specific musical features shaping listeners' perceptions. This study investigates the relationship between symbolic musical features and perceived emotional responses (valence and arousal) in Western classical music. Focusing on 72 preludes by J.S. Bach (Baroque period), F. Chopin (Romantic period), and D. Shostakovich (Modernist period), we develop an approach to predict emotional ratings directly from symbolic music that reduces interpretation bias, while examining how compositional structures shape emotional expression. It is observed that a GRU model integrated with Dyna-Octuple feature set achieves the best trade-off between accuracy and efficiency. Segment-level analysis reveals that emotional predictions remain stable across beginning, middle, and end sections, while period-based comparisons uncover distinctive stylistic patterns: Bach's balanced control, Chopin's expressive emotional intensity, and Shostakovich's contrasting emotional states. This preliminary study demonstrates how symbolic features in Western classical music can relate well to emotional responses, as well as providing insights into stylistic differences across different periods in history.

## CCS Concepts

• **Applied computing** → **Sound and music computing**; • **Computing methodologies** → *Feature selection.*

## Keywords

Symbolic music features, music emotion prediction, classical music analysis

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## 1 Introduction

Since ancient times, music has existed as a universal expression in delivering emotional influences among individuals beyond language. [4, 10, 32, 37] In the musical evolution, Western classical music with rich compositional structures and expressive markings continues to shape the harmonic, melodic, and dynamic patterns of modern popular music, demonstrating that music is a living art form that continually develops and adapts to the changing culture. [18] Therefore, examining the relationship between music and emotion through Music Emotion Recognition (MER), a sub-task of Music Information Retrieval (MIR)[32], provides a basis for understanding patterns of emotional expression and supports applications such as music recommendation[6] and retrieval [5].

In this study, we consider MER from the aspect of human perceived emotional attributes: *valence* and *arousal* [3]. Valence refers to the positivity or negativity of an emotion, while arousal indicates the intensity or level of excitement associated with emotions. Existing MER studies mostly rely on audio-based datasets such as EMMA [29] and MuSe [1], partly due to the ease of extracting information and features from audio data. A recent study [12] highlights the scarcity of publicly accessible datasets for Western classical music in the MER domain. In contrast, popular music, often freely accessible through online streaming platforms, is much more readily available for research.

We summarise the motivations of MER of Western classical music based on symbolic music data, which addresses two challenges:

**Challenge 1: Performance interpretation bias**: Audio recordings capture the performer's interpretation of the musical work, incorporating their own emotions and stylistic choices. These factors may diverge from, and potentially confuse, the composer's original emotional intent as expressed in the score [34]. It is therefore essential to encode the original composer's intended dynamics and expressions in the musical piece.

**Challenge 2: Limited interpretability:** Interpretation of music involves the message being delivered as an emotional sequence [39]. However, audio features offer limited explanatory power of how certain musical events correspond to specific emotional outcomes because raw audio inputs encode the acoustic signal directly but lack explicit representations of musically meaningful features such as melody and structure [24, 27], making it challenging to trace emotional responses back to specific musical events. Incorporating symbolic features can therefore improve interpretability by linking musical structures to emotional outcomes.

From these gaps, we underline two novel directions that are worth the attention of the MIR and multimedia community at large:

**Symbolic music for emotion prediction**: Symbolic representations, such as MIDI or **kern formats [26], encode more structural score-level information such as bar, tempo, and pitch, [38] while minimising the risk of performance bias. By extracting symbolic features (e.g., pitch range, dynamic level), the machine learning models can focus on the underlying musically meaningful compositional structure. This approach enhances interpretability by statistically linking these symbolic features to emotional dimensions.[40]

**Segment-level analysis**: Existing MER studies predict emotions from full or fixed-length pieces[19, 22]. Emotional expression often varies across different sections of a composition due to shifts in harmony, rhythm, or dynamics. To address this, we divide each piece into three structural segments: A (beginning), B (middle), and C (end) to conduct segment-wise prediction. This allows us to (i) identify which segment best aligns with the overall label and (ii) examine whether emotional cues are uniformly distributed or concentrated in specific positions [13, 35].

The contributions from this study can be summarised as follows:

- **Novel symbolic feature representation:** We propose *Dyna-Octuple*, a newly crafted symbolic representation that integrates the structural advantages of MusicBERT-Octuple with dynamic markings from **kern scores, replacing MIDI velocity values with score-derived dynamic intentions.
- **Systematic model comparison:** We provide a benchmark for recurrent neural networks (GRU, LSTM), tree-based models (Random Forest, XGBoost), and a large-scale pre-trained symbolic transformer (MusicBERT) on the MER task, evaluating predictive accuracy and training efficiency. Our experimental results (in Sec. 4) verify that GRU with Dyna-Octuple pair significantly outperforms the rest, having the lowest RMSE with a reasonably short training time.
- **Segment-level emotional consistency analysis:** We examine model performance across temporal segments (beginning, middle, end) of each piece, enabling assessment of whether emotional cues are uniformly distributed or concentrated in specific segments, thus offering further insights into symbolic MER.

To guide this work, we make the following assumptions:

(1) **Emotional ratings are consistent among raters.** The emotional labels sourced from peer-reviewed studies are reliable and accurately reflect the intended or perceived emotional content of the musical pieces.
(2) **Symbolic data reflects core musical elements.** Symbolic music data accurately represents the compositional structure and original emotional intent of the composer, limiting the risk from performance-specific variations.
(3) **Emotional ratings remain consistent throughout the piece.** The beginning, middle, and end segments of a single short piece (i.e. 'Prelude') share the same emotional label.

# 2 Related Work

## 2.1 Symbolic feature extraction and embedding

Automatic extraction of high-level descriptors from symbolic music has a long history in MIR. Music21 [8] enables parsing and analysis of symbolic data, while jSymbolic [21] extracts hand-crafted features such as tonality, melody, and rhythm from MIDI and kern [26]. Recent representation learning models such as MIDI2vec [20] and MusicBERT-Octuple [38] have leveraged neural embeddings of symbolic sequences that capture contextual pitch, timing, and dynamic information for downstream tasks.

## 2.2 Symbolic Datasets for MER

*2.2.1 Emotion-Labelled Symbolic Datasets.* Emotion-labelled symbolic datasets mainly cover popular music, with limited resources for Western classical music. EMOPIA provides pop-piano MIDI–audio pairs with clip-level valence/arousal labels [15]. XMIDI [30], the largest symbolic dataset with emotion and genre tags, includes only 11% classical pieces, limiting large-scale emotion recognition research in classical music.

*2.2.2 Emotion-Unlabelled Symbolic Datasets.* Large-scale unlabelled symbolic datasets are widely available for pre-training or feature extraction in MER. The GiantMIDI-Piano dataset provides over 10,800 unique classical solo piano works, without emotion labels [16]. while the Lakh MIDI dataset mainly covers pop and rock [25]. Despite lacking emotion annotations, these datasets may be valuable for unsupervised or self-supervised learning in this domain.

*2.2.3 Custom Feature Extraction from Limited Emotion-Labelled Data.* In the scarcity of emotion-labelled symbolic datasets, researchers extract custom features with toolkits like jSymbolic [21] and music21 [8], providing descriptors such as pitch range and rhythmic variation for MER. More recent approaches employ learned embeddings from symbolic transformer models (e.g., MusicBERT [38]) or domain-adapted formats like Dyna-Octuple, which capture fine-grained event sequences. Even with limited datasets, these features have demonstrated strong emotion recognition performance [15, 30], often comparable to audio-based models.

## 2.3 Model architectures for symbolic music and MER

*2.3.1 Tree-based methods.* Early MER approaches applied tree-based regressors (Random Forests, XGBoost) to symbolic features [17, 31, 36]. They are interpretable and data-efficient, but the lack of sequential modelling limits its performance.

*2.3.2 Recurrent Neural Networks (RNNs).* Recurrent neural networks (LSTM, GRU) effectively capture temporal dependencies in symbolic music data [2, 11, 14], offering a trade-off between accuracy and computational cost. Extensions with attention or hierarchical structures further enable RNNs to model both local patterns and long-range dependencies in symbolic sequences.

*2.3.3 Transformers.* Transformer-based pre-trained models have recently advanced symbolic MER. MusicBERT [38], using OctupleMIDI encoding and bar-level masking, learns transferable representations for emotion classification. Similarly, MidiBERT-Piano [7], pretrained on large-scale piano MIDI corpora, performs well across symbolic music understanding benchmarks, often surpassing RNN-based models. However, in sequence-level emotion classification, MusicBERT outperforms MidiBERT-Piano, suggesting that broad symbolic pre-training offers better generalization than domain-specific (piano-focused) pre-training for diverse MER tasks.

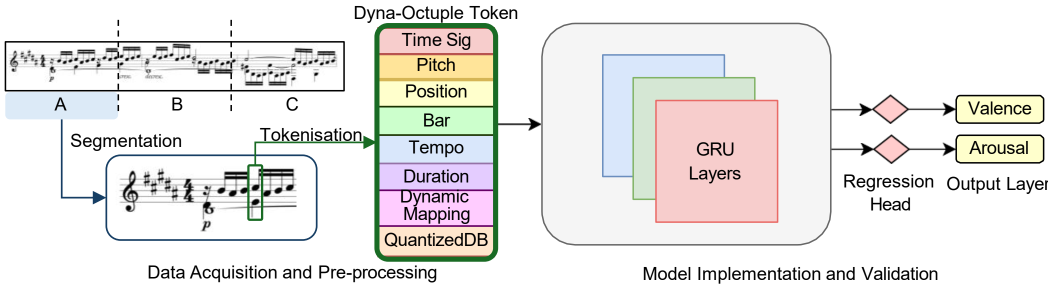


**Figure 1: Pipeline from data acquisition, segmentation into: A (beginning), B (middle), and C (end), and pre-processing of symbolic scores (Bach's Prelude No. 23 in the example), through feature representation and tokenisation into Dyna-Octuple tokens, to modelling and regression outputs for valence and arousal.**

# 3 Methodology

## 3.1 Data preprocessing

*3.1.1 Symbolic music dataset.* In this work, we use a symbolic piano music dataset of 72 preludes from three Western classical composers representing distinct periods: J.S. Bach (Baroque), F. Chopin (Romantic), and D. Shostakovich (20th-century/modernist). In Western classical tradition, a *prelude* is a short piece that typically sustains a single emotional affect. This characteristic is also found in the structure of modern pop songs [9, 28]. The dataset was sourced in both MIDI and Humdrum formats via the CCARH repository[1]. Dynamic markings in `**kern` data format were manually annotated using the Verovio Humdrum interface[2]. Each dynamic marking was mapped into decibel (dB SPL) following the Humdrum documentation [26], and then normalised into 32 discrete values for consistent representation across the dataset [38].

*3.1.2 Emotional Labels.* For each prelude, the ground truth emotional labels are continuous valence and arousal ratings obtained from a peer-reviewed source [9] with values rescaled to a canonical [0,1] range using min-max normalisation. Each piece is segmented into three equal-duration parts denoted as: A (beginning), B (middle), and C (end). Each segment inherits the piece-level emotion label under the assumption that the emotional ratings are similar.

## 3.2 Feature extraction and selection

In this section, we first elaborate the proposed *Dyna-Octuple* features, followed by brief description of the two comparative feature sets: the MusicBERT-Octuple [38] and jSymbolic-Octuple.

*3.2.1 Dyna-Octuple.* To leverage on score dynamics, we directly incorporate this information by manually extracting the dynamics directly from the preprocessed `**kern files`. The proposed **Dyna-Octuple** consists of the following elements (see Figure 2):

(1) **Time_signature** ($\mathcal{S}$): The numerator of the time signature, 4/4 time by defaults.
(2) **Tempo** ($\mathcal{T}$): Microseconds per quarter note. 500,000 microseconds per quarter note, which corresponds to 120 beats-per-minute (BPM).
(3) **Bar**: The measure (or bar) number of the note, calculated as:

$$\text{bar} = \left\lfloor \frac{t_{start}}{d_{perbar}} \right\rfloor + 1 \quad (1)$$

$t_{start}$ is the start time of the note in seconds, and the result is incremented by 1 to avoid starting from zero. The duration per bar (in seconds) is computed as:

$$d_{perbar} = \left(\frac{60}{\mathcal{T}}\right) \times \mathcal{S} \quad (2)$$

where the time per beat (i.e. quarter note) is multiplied by $\mathcal{S}$ to give the total time for a bar.
(4) **Position**: The position of the note within its bar (measure), calculated as the remainder when dividing the note's start time by the seconds per bar:

$$\text{position} = t_{start} \ \% \ d_{perbar} \quad (3)$$

(5) **Pitch**: Corresponds to the MIDI pitch number in the range of [1,128]. (e.g., 60 = Middle C in the piano roll).
(6) **Duration**: The duration of the note, calculated as:

$$\text{duration} = t_{end} - t_{start} \quad (4)$$

where note.end is the end time of the note, and note.start is the start time of the note in seconds.
(7) **DynamicMapping**: Dynamic markings (e.g. *'p'* for soft, *'ff'* for very loud) extracted from `**kern` are translated to dynamic levels expressed in decibels, according to the official Humdrum conversion mapping[3].
(8) **QuantizedDB**: Dynamic markings extracted from the MIDI key-velocity data are translated to an estimated decibel value, and then mapped to a set of 32 discrete levels. This ensures a standardised representation of intensity across different musical compositions.

*3.2.2 MusicBERT-Octuple.* From the symbolic encoding of the MIDI file's events, the MusicBERT-Octuple [38] is structurally formed with the following extracted features:

(1) **Measure**: Measure of the structural position of the note within the composition.
(2) **Position**: The position of the note in the current measure indicating the exact position of a note in relation to the beats.

[1]https://kern.ccarh.org/
[2]https://verovio.humdrum.org/
[3]https://www.humdrum.org/guide/ch28/#the-db-command

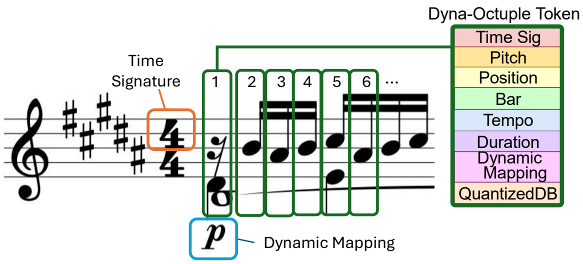


**Figure 2: Dyna-Octuple feature extraction from MIDI and **kern scores. All notes are tokenised vertically at each time step into Dyna-Octuple tokens, ensuring complete event-level granularity. (Bach's Prelude No. 23 in the example)**

(3) **Program**: Represents the instrument that produces the note, given by the MIDI program number.
(4) **Pitch**: The MIDI pitch value of the note in range [1,128].
(5) **Duration**: Represented in terms of MIDI ticks to indicate how long the note is held.
(6) **Velocity**: Represents note intensity, derived from MIDI velocity, quantised into 32 values ranging from 2 to 126.
(7) **Time Signature**: The time signature during the note, showing beats per measure and beat type.
(8) **Tempo**: The tempo of the piece at the time of the note, measured in beats per minute (BPM).

*3.2.3 jSymbolic-Octuple.* jSymbolic-Octuple is derived from the open-source jSymbolic toolkit, a Java-based application that extracts 246 unique symbolic music features from MIDI files [21]. For consistency in our experimentation, we select the top 8 descriptors with the highest Pearson correlation to valence and arousal labels. The features of jSymbolic-Octuple are as follows:

(1) **Rhythmic Variability**: Measures the diversity of note durations, capturing rhythmic complexity.
(2) **Rhythmic Variability (Tempo-Standardised)**: Same as above, but normalised by tempo to handle speed differences.
(3) **Importance of High Register**: Quantifies how prominently high-pitched notes are used in the piece.
(4) **Mean Pitch**: The average pitch across all notes, reflecting the general register or brightness of the music.
(5) **Average Length of Melodic Arcs**: Captures how long melodic phrases tend to extend before changing direction.
(6) **Most Common Rhythmic Value**: The most frequently occurring rhythmic duration (e.g., eighth note, quarter note).
(7) **Prevalence of Most Common Pitch**: Indicates how dominant the most repeated pitch is within the piece.
(8) **Combined Strength of Two Strongest Rhythmic Pulses (Tempo-Standardised)**: Measures the salience of dominant beat patterns, normalised by tempo.

## 3.3 Model Implementation

In this work, we implement five supervised models to evaluate the three proposed feature representations across musical segments.

- **MusicBERT** [38]: A pretrained transformer, fine-tuned with a double linear regression head (valence and arousal).

**Table 1: Input shapes of different models/feature sets used**

| Model | Feature Set | Input Shape |
|---|---|---|
| MusicBERT | Dyna-Octuple | (512, 8) |
| | MusicBERT-Octuple | (512, 8) |
| LSTM / GRU | Dyna-Octuple | (512, 8) |
| | MusicBERT-Octuple | (512, 8) |
| | jSymbolic-Octuple | (128, 8) |
| RF / XGB | jSymbolic-Octuple | (1024) |

- **Long Short-Term Memory (LSTM):** A bidirectional LSTM architecture with two layers comprising 50 and 25 hidden units, respectively, is employed to capture temporal dependencies in the sequential input. The output is passed through two dense layers: one with 10 units using ReLU activation, followed by a linear output layer of size 2.
- **Gated Recurrent Unit (GRU):** The GRU model followed the same structure as the LSTM, with two layers of 50 and 25 hidden units, respectively, and the same dense output layers.
- **Random Forest (RF):** Inputs are flattened and standardised using z-score normalisation. Hyperparameters such as maximum tree depth and the number of estimators are optimised through internal cross-validation.
- **XGBoost (XGB):** Inputs are flattened and standardised in the same manner as RF. Key parameters, including learning rate, maximum depth, and number of estimators, are tuned to optimise performance.

For the sequential models, the input sequence is defined as:

$$S = \{s_1, s_2, \ldots, s_T\}, \quad s_t \in \mathbb{R}^8, \quad S \in \mathbb{R}^{T\times 8} \tag{5}$$

where $s_t$ denotes the 8-dimensional (octuple) vector at timestep $t$. Each note is tokenised vertically at each time step. The sequence length $T$ is standardised by truncation or zero-padding to match the required input size.

For the tree-based models (RF and XGB), the sequence is flattened into a one-dimensional vector:

$$\text{Flatten}(S) = \{s_1^{(1)}, s_1^{(2)}, \ldots, s_1^{(8)}, s_2^{(1)}, \ldots s_T^{(8)}\} \in \mathbb{R}^{T\times 8} \tag{6}$$

The sequence length and dimensionality are shown in Table 1.

Sequential models (LSTM and GRU) were implemented in TensorFlow/Keras and trained with the Adam optimiser (clipnorm = 1.0), a batch size of 16, and early stopping after three epochs without improvement. Tree-based models (RF and XGB) were implemented using scikit-learn and other related libraries. All experiments were conducted with an NVIDIA GeForce RTX 3050. The extracted features and emotion annotations used in this study will be released at https://github.com/sui-weien/Dyna-Octuple.

## 3.4 Evaluation Metrics

The following regression metrics are used to evaluate model performance during validation:

- **Mean Squared Error (MSE)**: Emphasises larger errors by squaring the residuals, making it sensitive to outliers and useful when large deviations are undesirable.

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{7}$$

**Table 2: Performance evaluation across different musical segments (A, B, C) averaged across all models, with the segment as the manipulated variable. The best-performing model for each metric is highlighted in bold.**

| Segment | MSE ↓ | MAE ↓ | RMSE ↓ |
|---|---|---|---|
| A | 0.0429 | **0.1756** | 0.2061 |
| B | 0.0434 | 0.1768 | 0.2073 |
| C | **0.0427** | 0.1760 | **0.2058** |

**Table 3: Performance comparison of all models used in this study. Results are averaged across the three musical segments (A, B, C) and across all applicable feature sets for each model.**

| Model | MSE | MAE | RMSE |
|---|---|---|---|
| MusicBERT | 0.0462 | 0.1818 | 0.2136 |
| LSTM | 0.0432 | 0.1760 | 0.2068 |
| Random Forest | 0.0423 | 0.1752 | 0.2049 |
| XGBoost | 0.0423 | 0.1751 | 0.2048 |
| GRU | **0.0410** | **0.1725** | **0.2018** |

- **Mean Absolute Error (MAE)**: Provides a linear and interpretable measure of average error magnitude, treating all errors equally regardless of direction or size.

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{8}$$

- **Root Mean Squared Error (RMSE)**: Expresses the error in the same unit as the target variable, making it easier to interpret, while still penalizing larger errors more than MAE.

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{9}$$

# 4 Results and Analysis

## 4.1 Segment Comparison

For each prelude, we evaluated model performance across three segments—A (beginning), B (middle), and C (end). Table 2 shows that all three segments achieved similar RMSE values with pairwise t-tests between segments yielding $p$-values above 0.6, indicating no significant differences. Thus, the following experiments would report results averaged over all three segments.

These findings suggest two observations: (1) Models' performance remains consistent across segments; (2) The emotionally predictive content is evenly distributed across a prelude piece, with no single segment holding a dominant influence on emotion prediction. This aligns with the structural characteristics of preludes, which are short piano works typically composed to sustain one dominant affect.

## 4.2 Models Comparison

By taking the average performance over all three segments, Table 3 compares the performance of various models on MSE, MAE and RMSE metrics. Among them, GRU achieved the lowest RMSE (0.2018), alongside marginal improvements in MSE and MAE.

A one-way ANOVA on RMSE values stated a statistically significant difference in performance across models ($F = 6.292$, $p = 0.0027$). Pairwise t-tests further indicated the GRU outperformed MusicBERT, Random Forest and XGBoost at the $p < 0.05$ significance level. However, no significant difference was observed between LSTM and all other models ($p > 0.05$). The GRU outperforms non-recurrent models by capturing temporal patterns in symbolic music while avoiding overfitting on small datasets. Its simple architecture offers a balance between accuracy and training efficiency.

**Table 4: Performance comparison of different feature representations used in this study. Results are averaged across all models that can process each feature set.**

| Feature Category | MSE | MAE | RMSE |
|---|---|---|---|
| Jsymbolic-Octuple | 0.1460 | 0.3160 | 0.3512 |
| Dyna-Octuple | 0.0673 | 0.2086 | 0.2417 |
| MusicBERT-Octuple | **0.0434** | **0.1770** | **0.2073** |

## 4.3 Feature Set Comparison

Table 4 compares the performance of the three feature sets on MSE, MAE, and RMSE metrics, averaging across all models compared in Sec. 4.2. Among all feature sets, MusicBERT-Octuple achieved the lowest RMSE (0.2073), followed by Dyna-Octuple (0.2417). However, its advantage over the Dyna-Octuple was not significant.

A one-way ANOVA stated a significant difference among the feature sets ($F = 4.920$, $p = 0.0151$). Pairwise t-tests showed that MusicBERT-Octuple significantly outperformed Jsymbolic-Octuple ($t = 3.258$, $p = 0.0076$), confirming its stronger emotion representational ability. In contrast, the difference between Dyna-Octuple and MusicBERT-Octuple was not statistically significant ($t = 1.068$,$p = 0.3166$), suggesting that the score-derived dynamics contribute comparably to prediction accuracy. The comparison between Dyna-Octuple and Jsymbolic-Octuple was marginally non-significant ($p = 0.0597$). Still, the overall observation indicates that incorporating dynamics from the score brings more informative emotional elements than purely symbolic descriptors.

## 4.4 Feature-Model Pair Comparison

To better understand the effect of encoding score dynamics into the feature set and to precisely uncover the best feature-model pairing, we limit the feature sets to vary between MusicBERT-Octuple and Dyna-Octuple while using different models to learn. Table 5 reports that the best performance was achieved by GRU with Dyna-Octuple features (RMSE = 0.2011), closely followed by GRU with MusicBERT-Octuple (RMSE = 0.2026). By contrast, MusicBERT trained on its own features performed worst with the highest error (RMSE = 0.2154). These results suggest that dynamic markings from `**kern` can provide a more precise dynamic representation of expressive intent, potentially offering richer emotional cues than the MIDI velocity encoding used in MusicBERT-Octuple.

We outline the following observations: (1) GRU consistently outperforms LSTM and MusicBERT across feature sets, suggesting that the GRU is better suited for this task. Its simpler architecture and efficient gating mechanisms are well-aligned with the nature of the prelude dataset which contains shorter emotion dependencies, allowing it to capture emotional cues effectively while reducing training time. (2) Dyna-Octuple is particularly effective when paired

**Table 5: Performance comparison of feature-model pairs. Each row corresponds to a combination of feature representation and model**

| Feature Set | Model | MSE | MAE | RMSE |
|---|---|---|---|---|
| MusicBERT-Octuple | MusicBERT | 0.0471 | 0.1822 | 0.2154 |
| Dyna-Octuple | LSTM | 0.0452 | 0.1777 | 0.2113 |
| Dyna-Octuple | MusicBERT | 0.0448 | 0.1812 | 0.2110 |
| MusicBERT-Octuple | LSTM | 0.0419 | 0.1749 | 0.2038 |
| MusicBERT-Octuple | GRU | 0.0413 | 0.1737 | 0.2026 |
| Dyna-Octuple | GRU | **0.0407** | **0.1712** | **0.2011** |

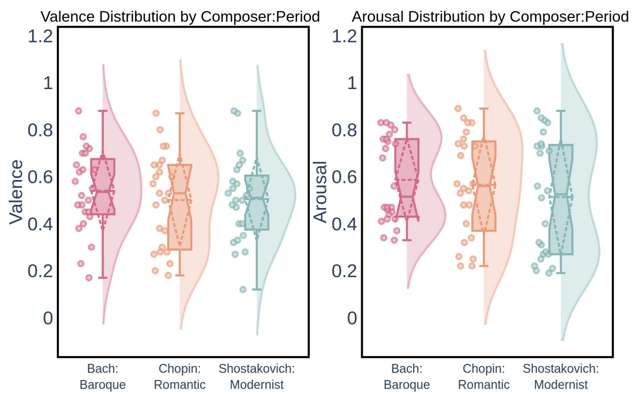


**Figure 3: Valence and Arousal Distribution by Period**

with GRU, suggesting score-derived dynamic information is more effective than direct embeddings from symbolic music.

## 4.5 Valence and Arousal Distribution by Period

The composite dataset comprising preludes from three different composers presents an opportunity to gain insights into the distribution of emotional dimensions by period (see Figure 3).

**Bach (Baroque):** Music from the Baroque period is characterised by formal structures with order. While Bach's music encompasses a wide range of emotions, the expression might be perceived as more controlled and less overtly sentimental. The *unimodal* violin plot for Valence distribution centered at 0.5 reflects the balance of emotional content. The *bimodal* Arousal distribution reflects two distinct styles typical of Baroque music— one that is more virtuosic and expressive, another more stately and mellow.

**Chopin (Romantic):** Music from the Romantic period emphasised emotional expression, subjectivity with the expressive use of lyrical melodies, dynamics and rubato. The Valence plot showed a tendency towards higher values and a wider spread, which aligns with the more emotional nature of Romantic music. The Arousal distribution for Chopin also appears as a long-tailed curve, covering a broader range of emotional intensity than the Baroque pieces.

**Shostakovich (Modernist):** Music from the Modernist era explored new harmonic combinations, rhythmic complexities, with a broader spectrum of emotional expression. The Valence distribution appeared *unimodal* with a wide spread, reflecting an overall balance but greater variability in emotional content. In contrast, the *bimodal* Arousal distribution suggests that some works are depressed and passive, while others are active and intense.

## 4.6 Summary of Findings

This preliminary work investigates symbolic music emotion recognition (MER) on Western classical music and how dynamics-driven feature representation coupled with a suitable learning model produces reliable predictions. Segment-level analysis showed that predictive accuracy was consistent across all parts of the preludes dataset, supporting the assumptions made. Comparative experiments demonstrated that GRU combined with the Dyna-Octuple achieved the best trade-off between accuracy and efficiency, outperforming LSTM and MusicBERT. Finally, the period-based analysis revealed distinct emotional patterns: Bach's music showed balanced Valence and bimodal Arousal, Chopin's works displayed greater emotional intensity, and Shostakovich's music exhibited the widest variability in emotions.

# 5 Conclusion

This study shows that symbolic music emotion recognition (MER) can be effectively advanced through expressive symbolic music representations and well-matched selected model architectures. By introducing Dyna-Octuple, which integrates score-derived dynamic markings, we improved interpretability and captured dynamic intention directly from the composer beyond conventional MIDI features. The result shows that GRU with Dyna-Octuple provides the best balance between accuracy and efficiency, outperforming larger pre-trained models such as MusicBERT under the current dataset. Furthermore, segment-level analysis showed consistent predictions across preludes, consistent with their nature as short works built around a single affect. Cross-era comparisons highlighted distinctive stylistic patterns: Bach's balanced control, Chopin's expressive emotional intensity, and Shostakovich's contrasting states of passivity and activity. Overall, these findings highlight the importance of symbolic feature design, model efficiency, and stylistic context in advancing MER research, particularly within the underexplored domain of Western classical music.

*Limitations.* This study is constrained by the limited size and scope of the prelude dataset, which may reduce the generalizability of results to other musical forms, cultural contexts, and genres. In addition, the subjectivity of emotion annotation introduces potential biases to the learned model.

*Future directions.* In future, we foresee our work being expanded to larger and more diverse symbolic music datasets to improve robustness and generalizability across different styles and cultures. Developing personalised emotion recognition models that incorporate contextual factors (e.g., mood, environment) may further enhance relevance and precision [23]. Additionally, exploring domain-adaptive pre-training strategies for symbolic music transformers and real-time emotion recognition applications could further advance the field [33].

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